



**BMS COLLEGE OF ENGINEERING, BENGALURU-19**  
Autonomous Institute, Affiliated to VTU  
**DEPARTMENT OF MATHEMATICS**

|                          |                                                  |                   |                                     |                   |            |
|--------------------------|--------------------------------------------------|-------------------|-------------------------------------|-------------------|------------|
| <b>Sem &amp; Branch:</b> | Third Semester<br>CSE, ISE, CS-DS, CS-IOT, AI-DS | <b>Subject:</b>   | STATISTICS AND DISCRETE MATHEMATICS | <b>Sub Code:</b>  | 23MA3BSSDM |
| <b>Time:</b>             | 12:45 PM-2:00 PM                                 | <b>Test Date:</b> | 04/11/2024                          | <b>Max Marks:</b> | 40         |

| Test No. | Q. No. |  | Marks | CO |
|----------|--------|--|-------|----|
|----------|--------|--|-------|----|

- i) Questions from 1 to 5 are compulsory.  
ii) Answer either Question 6a or 6b.

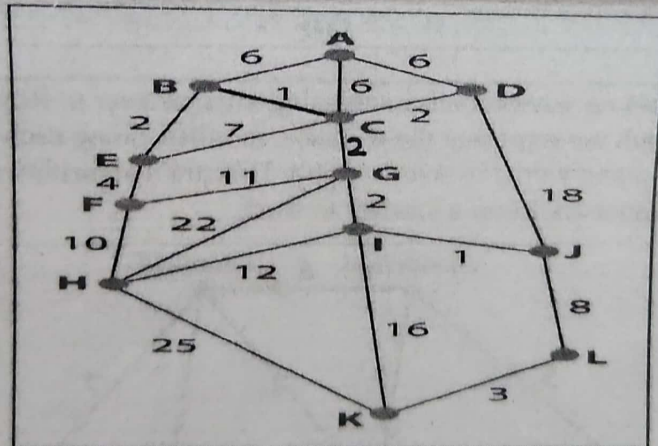
- 1 a) A sports tournament has 12 teams, and each team plays against every other team exactly once. How many total matches were played?  
b) Find the number of edges in a graph  $G$  with 6 vertices, where one vertex has a degree of 4, two vertices have a degree of 3, and the rest have a degree of 2.  
b) Draw the graph  $K_4$  and hence draw its spanning subgraph.

6  
1

- 2 On your trip to Venice, you plan to visit all the important world heritage sites but are short on time. To make your itinerary work, you decide to use Kruskal's algorithm. The table below indicates the important locations on the map and the edge weight represents the distance in metres. Obtain the minimal spanning tree of the given network.

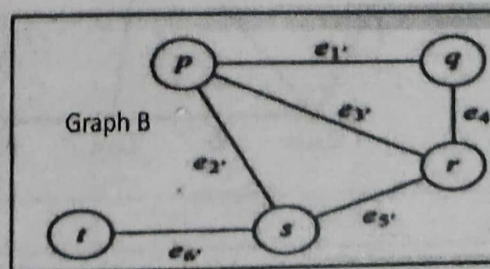
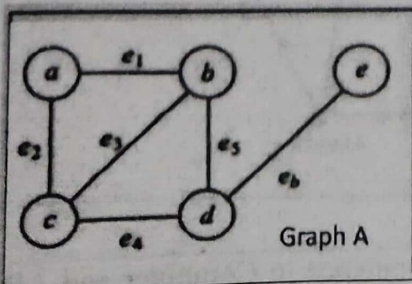
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|              |   |               |   |            |   |                   |   |
|--------------|---|---------------|---|------------|---|-------------------|---|
| Cannaregio   | A | Dell 'Orto    | D | San Polo   | G | St. Mark Basilica | J |
| Ponte Scalzi | B | Ferrovia      | E | Dorso Duro | H | Castello          | K |
| Santa Corce  | C | Piazzale Roma | F | San Marco  | I | Arsenale          | L |



- 3 Define isomorphism. Check whether the following graphs are isomorphic or not.

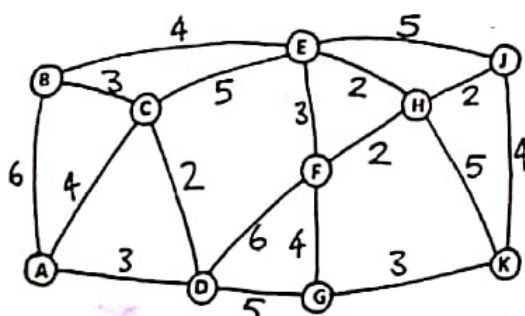
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|                 |                                                                                                      |                                                                                                                                                                                                                                                                                                  |   |   |
|-----------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| TEST-1          | 4                                                                                                    | A social media platform wants to analyze user connections to identify influential users. User A has friendships with B, C, D and E; B has friendships with C and D, while C has friendship with D. Represent this connection as a graph and hence write its adjacency and incidence matrix.      | 7 | 1 |
|                 | 5                                                                                                    | Suppose a committee has seven members, these members meet each day for lunch at a round table. They decide to sit in such a way that every member has different neighbors at each lunch. Explain using graph theory in how many ways can this arrangement last and also list all possible cases. | 7 | 1 |
|                 | 6a                                                                                                   | <p>Consider the map below. The cities have been selected and marked from alphabets A to F and every edge has a cost associated with it. We need to travel from <b>Bengaluru(B)</b> to all other places. Identify the shortest paths with minimal cost from Bengaluru to other destinations.</p>  | 7 | 1 |
|                 | OR                                                                                                   |                                                                                                                                                                                                                                                                                                  |   |   |
|                 | 6b                                                                                                   | <p>The diagram below shows roads connecting villages near to Royton. The numbers on each arc represent the distance, in miles, along each road. Leon lives in Royton and works in Aston. Apply Dijkstra's algorithm to find the minimum distance for Leon's journey to work.</p>                 | 7 | 1 |
| Course Outcome: |                                                                                                      |                                                                                                                                                                                                                                                                                                  |   |   |
| CO 1            | Apply the concept of Discrete Mathematics and Statistics in Computer and Allied Engineering Science. |                                                                                                                                                                                                                                                                                                  |   |   |



|               |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                        |              |             |
|---------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------|-------------|
| Sem & Branch: | Third Semester<br>(CS, IS, DS, IOT) | Subject:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | STATISTICS AND DISCRETE<br>MATHEMATICS | Sub<br>Code: | 23MA3BSSDM  |
| Time:         | 09:00 AM-10:15 AM                   | Test Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 04/01/2024                             | Max Marks:   | 40          |
| Test<br>No.   | Q.<br>No.                           | ANSWER ALL QUESTIONS<br>Questions from 1 to 5 are compulsory.<br>Internal choice is provided in question 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |              | Marks<br>CO |
| 1             | 1                                   | Model the following situation as graph: Ada and Bertrand are friends. Ada is also friends with Cecilia and David. Bertrand, Cecilia and Lily are all friends of each other. Also, write the adjacency matrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |              | 6<br>1      |
| 2             | 2                                   | Determine the order of the graph in the following cases:<br>(i) G has 17 edges, 1 vertex of degree 2, 2 vertices of degree 3, 3 vertices of degree 4 and all other vertices of degree 7.<br>(ii) G is a regular graph with 24 edges. <i>every vertex has equal deg</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                        |              | 6<br>1      |
| 3             | 3                                   | Prove that a connected graph G remains connected even after removing an edge e from G if and only if e is a part of some cycle in G.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                        |              | 7<br>1      |
| 4             | 4                                   | <p>Once upon a time there was a city that had no roads. Getting around the city was particularly difficult after rainstorms because the ground became very muddy. Cars got stuck in the mud and people got their boots dirty. The Mayor of the city decided that some of the streets must be paved, but didn't want to spend more money than necessary because the city also wanted to build a swimming pool. The Mayor therefore specified two conditions:</p> <ol style="list-style-type: none"> <li>1. Enough streets must be paved so that it is possible for everyone to travel from their house to anyone else's house only along paved roads, and</li> <li>2. The paving should cost as little as possible.</li> </ol>  <p>The layout of the city is represented by graph given above.<br/>Apply Kruskal's algorithm to find the best route that connects all the houses.</p> |                                        |              | 7<br>1      |
| 5             | 5                                   | Let G be a simple graph with n vertices and m edges where n is at least 3. If $m \geq \frac{(n-1)(n-2)}{2} + 2$ , prove that G is a Hamiltonian graph.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                        |              | 7<br>1      |

ii) By H.S,  $\sum \deg(v) = 2|E|$   
 $x = 2 \times 24$

Order =  $x = 48$

(i)  $1 \times 2 = 2$   
 $2 \times 3 = 6$   
 $3 \times 4 = 12$   
 $2 \times 7 = 14$

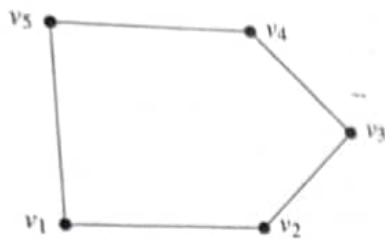
Vertices =  $1 + 2 + 3 + 2 = 8$   
 $20 + 17 = 37$   
 $37 - 20 = 17$   
 $17 \times 2 = 34$   
 $34 - 20 = 14$   
 $14 \times 2 = 28$



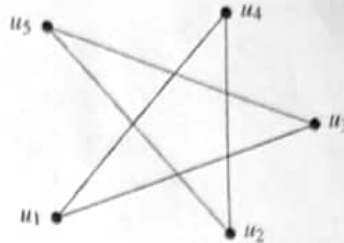
is all isomorphic  $\rightarrow$  (i) same no. of vertices  
 $C_1$  edges  
 (ii) equal no. of vertices with a given degree

a) Check whether the following graphs are isomorphic or not:

6



(a)  $G_1$ .

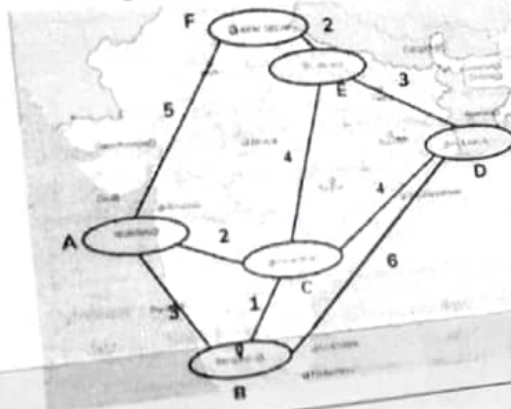


(b)  $G_2$ .

OR

b) Consider the map below. The cities have been selected and marked from alphabets A to F and every edge has a cost associated with it. We need to travel from **Bengaluru(B)** to all other places. Identify the shortest paths with minimal cost from Bengaluru to other destinations.

7



**Course Outcome:**

**CO 1**

Apply the concept of Discrete Mathematics and Statistics in Computer and Allied Engineering.

